

Robust Nighttime Vehicle Localization Under Headlight Saturation via Ensemble Detection and Actuation-Aware 3-D Tracking

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Abstract

Nighttime vehicle localization becomes particularly fragile when intense headlights, scattering, and abrupt exposure changes degrade the visual signal. In camera-centric adaptive driving beam (ADB) systems, such degradation can cause unstable or missing detections and consequently lead to incorrect beam-suppression decisions. This paper presents AuraBeam, an application-oriented machine-vision framework for analyzing closed-loop ADB behavior under nighttime visual degradation. AuraBeam integrates an ensemble YOLO detector, a physics-based Pseudo-Radar depth prior, a six-state 3-D Kalman filter, and an Arduino-based 8×8 LED Hardware-in-the-Loop platform. To evaluate downstream control quality, we use the Glare Suppression Success Rate (GSSR), an LED-grid metric designed to measure suppression fidelity beyond conventional pixel-space detection scores. Experiments are conducted on a pre-specified five-case protocol covering negative controls, targeted degradation conditions, and a limitation case. The controlled comparison between fixed 3-D observation and an adaptive observation fallback shows that the adaptive mechanism is not uniformly superior: it preserves performance in the control cases, matches the fixed-observation baseline in foggy-night and lightning-disturbance scenarios, and degrades performance in the rainy-night limitation case. These findings demonstrate that fixed 3-D tracking with heavy filtering often provides stronger

actuation stability, positioning adaptive switching as a scenario-dependent boundary case rather than a universal replacement. More broadly, the study demonstrates that actuation-aware evaluation is necessary for exposing perception failures that are not fully characterized by image-space detection metrics alone.

Keywords: adaptive driving beam, nighttime vehicle localization, hardware-in-the-loop, Kalman filtering, glare suppression

1 Introduction

Nighttime driving constitutes a disproportionately hazardous operating condition for road users. Although nighttime traffic volume is substantially lower than its daytime counterpart, severe crashes occur more frequently at night, and glare from oncoming headlights is widely recognized as a major contributor to degraded visual performance [1–3]. This challenge has motivated the development of intelligent front-lighting systems that seek to preserve the host driver’s visibility while reducing glare exposure for other road users. In particular, *Adaptive Driving Beam* (ADB) systems selectively suppress only the portion of the high beam that would dazzle an oncoming vehicle, thereby retaining as much useful illumination as possible outside the affected zone [4, 5]. From a computer-vision perspective, however, ADB is not merely a lighting problem. It is a closed-loop perception-and-actuation task in which the system must first localize the relevant oncoming road user in the image, then translate that localization into a beam-suppression pattern on a discrete headlight actuator. This coupling introduces a practical failure mode that remains insufficiently addressed in the literature: when the front-facing camera is exposed to intense oncoming headlights, blooming, flare, specular reflection, and rapid auto-exposure adaptation can severely degrade the visual signal [6, 7]. In the most adverse cases, the detector output becomes unstable or vanishes entirely, causing the suppression region to flicker or disappear altogether. As a result, a camera-centric ADB pipeline may fail precisely in the regime where robust localization is most safety-critical.

This limitation highlights a broader gap between mainstream object detection and the requirements of perception-guided beam control. Recent nighttime vehicle detection studies have demonstrated that low-light driving scenes differ fundamentally from daytime imagery: salient visual evidence is often concentrated in sparse, high-intensity headlight regions rather than in stable object texture or contour structure. Accordingly, recent approaches have explored specialized feature design, headlight-aware modeling, and highlight-guided attention to improve nighttime vehicle detection under challenging illumination conditions [8–10]. At the same time, the automotive perception literature has increasingly demonstrated the value of radar–camera fusion for maintaining robust localization when vision alone becomes unreliable, particularly under adverse weather, darkness, and low-visibility conditions [11–14]. Yet these two strands of work are rarely brought together in the specific context of ADB, where the goal is not general scene understanding, but stable continuation of the suppression

zone during brief periods of visual collapse. A second limitation of existing work is methodological. Most vision papers evaluate nighttime detection in pixel space using standard metrics such as Precision, Recall, and mAP [8, 15, 16], whereas automotive lighting studies emphasize photometric or regulatory performance. For perception-driven matrix headlight control, neither perspective alone is fully sufficient. Small image-space localization errors may map to incorrect LED cells, producing either residual glare or unnecessary darkening. Moreover, image-level detection scores do not directly capture whether a downstream suppression actuator behaves correctly during transient detector dropout. This motivates the need for an evaluation protocol that links perception robustness to actuation fidelity in a reproducible closed-loop setting.

To address these gaps, we propose **AuraBeam**, a low-cost, proof-of-concept application-oriented machine-vision framework for studying nighttime vehicle localization and glare suppression in camera-centric ADB under headlight saturation. Rather than treating nighttime detection as an isolated image-space task, AuraBeam combines detection, adaptive tracking, and physical actuation into a unified pipeline. First, an ensemble vision module executes YOLOv5 and YOLOv8 [17] in parallel and fuses their predictions using Weighted Non-Maximum Suppression [18] to obtain a more stable perception front-end under diverse nighttime conditions. Second, a six-state 3-D Kalman filter integrates the image-space observations with a physics-based depth surrogate (Pseudo-Radar) and can switch to a depth-prior-only observation mode during camera degradation [19], allowing the suppression trajectory to be propagated through short intervals of visual dropout. Third, the framework is validated in a compact Hardware-in-the-Loop (HIL) environment and evaluated using a new actuation-level metric, the *Glare Suppression Success Rate* (GSSR), which measures suppression fidelity on the LED grid rather than exclusively in pixel space. To this end, the main contributions of this work are as follows:

- **A reproducible closed-loop ADB research stack for glare degradation studies.** We design a dual-model YOLO front-end in which YOLOv5 and YOLOv8 are executed in parallel and fused by Weighted Non-Maximum Suppression, then coupled with Pseudo-Radar depth, 3-D tracking, and LED-grid actuation in a single Hardware-in-the-Loop pipeline.
- **Adaptive 3-D tracking analysis under temporary visual collapse.** We introduce a six-state 3-D Kalman filtering framework that fuses image-space detections with a physics-based depth surrogate and can switch its observation model when the camera signal becomes unreliable. The final study isolates this switching mechanism by comparing a fixed-observation variant against an adaptive variant under a pre-specified protocol, including both matched degradation cases and a rain-induced boundary case.
- **Actuation-aware evaluation in a closed-loop HIL setting.** We formulate the *GSSR* as an LED-grid-based metric for evaluating downstream suppression quality, and we construct a reproducible five-case evaluation protocol that explicitly separates negative controls, targeted degradation cases, and a limitation case.

Overall, this work aims to bridge three research areas that are typically studied in isolation: nighttime vehicle detection, robust tracking and fusion under degraded sensing,

and intelligent headlight actuation. The contribution is positioned as an engineering-oriented machine-vision application rather than a general-purpose detector benchmark. By focusing on the specific failure mode of *glare-induced detection collapse*, we show that camera-centric ADB should be evaluated not only through detection quality, but also through localization continuity and actuation-aware downstream behavior.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature and positions the proposed work within recent advances in nighttime vehicle detection, sensor fusion, and intelligent headlight systems. Section 3 details the proposed AuraBeam framework. Section 4 presents the evaluation protocol, datasets, and performance metrics. Section 5 reports and analyzes the experimental results. Finally, Section 6 summarizes the main findings and outlines directions for future work.

2 Background and Related Work

Adaptive and intelligent headlight systems have evolved from simple Auto High Beam (AHB) [20] switching toward spatially selective beam shaping that preserves forward visibility while reducing glare exposure for other road users. A recent review by Nkrumah *et al.* identifies camera-based scene understanding, rule-based beam modulation, and adaptive intensity control as the dominant paradigm in intelligent headlight systems [21]. In parallel, recent optical and regulatory studies have emphasized that production ADB/AFS designs must satisfy not only visibility requirements but also strict glare-compliance constraints under ECE-R123-like frameworks [22]. These developments confirm that modern headlight control is no longer a purely mechanical or optical problem; it is increasingly a perception-and-control problem in which the quality of the front-facing vision stack directly determines beam-selection safety. Despite this progress, most deployed and prototype ADB pipelines remain fundamentally *camera-centric*: an oncoming vehicle is first detected in the image plane, after which a suppression mask or beam cut-off region is generated for a segmented LED or pixelated headlamp. This architecture performs well under moderate nighttime conditions, but becomes fragile when the camera is saturated by strong oncoming headlights, specular reflections, or abrupt exposure changes. A recent literature review on vehicle-lighting glare further reinforces that glare remains an unresolved practical safety issue influenced by light source intensity, headlamp aim, road geometry, vehicle height, and retrofitted lighting configurations [3]. Thus, while recent literature has improved the design of adaptive beam systems, a key systems-level gap remains: how to maintain glare suppression when the vision channel itself temporarily collapses.

Nighttime vehicle detection is substantially more challenging than daytime detection because the scene is dominated by sparse, high-intensity light sources rather than stable object texture [23]. In such conditions, blooming, flare, reflection, and exposure adaptation distort the apparent spatial extent of the headlights and often erase the geometric contours of the vehicle body [7]. Recent work has therefore moved beyond generic daytime detectors toward methods that explicitly exploit nighttime-specific appearance cues [8–10]. For example, Zhang *et al.* proposed an improved HOG-based framework for on-road nighttime vehicle detection and tracking, demonstrating that

handcrafted structural cues can still be effective when illumination is poor and vehicle contours are weak [8]. More recently, Lashkov *et al.* developed a learning-based pipeline that couples headlight extraction with low-illumination tracking, further highlighting that vehicle lights themselves become the dominant visual carrier in dark scenes [9]. Deep models have also begun to incorporate this observation explicitly. HighlightNet, introduced by Song *et al.* [10], formulates nighttime vehicle detection as a dual-branch problem that jointly reasons about vehicle presence and highlight regions, using highlight-guided attention to direct the detector toward the most informative luminous structures. These recent studies collectively suggest an important shift in the literature: nighttime detection should not be treated as a simple low-light variant of daytime detection, but as a distinct visual regime in which the detector must reason under partial loss of texture, extreme local saturation, and ambiguous light-source morphology. However, most of these methods still evaluate performance primarily in pixel space, using standard detection metrics such as Precision, Recall, or mAP. For ADB control, this is only part of the problem. A detector may remain acceptable in pixel space while still producing unstable or misaligned suppression decisions once mapped onto a discrete LED grid. Moreover, many nighttime detection papers focus on *detecting* vehicles, whereas ADB requires *persistent localization* across short intervals of visual degradation. This distinction motivates our use of a dual-model detector and an actuation-aware downstream evaluation, rather than relying on detection performance alone.

Tracking plays a central role whenever detections are intermittent, noisy, or partially missing. Classical state estimators such as the Kalman filter (KF) [24] remain attractive because they provide a lightweight and interpretable mechanism for propagating object state through short periods of observation loss. Yet, in automotive perception, the more important recent trend is not simply filtering, but *multimodal fusion*. A comprehensive review by Yao *et al.* shows that radar-camera fusion has become a major paradigm for autonomous-driving perception because cameras provide rich semantics while radar offers depth and motion cues that are far more robust to darkness, adverse weather, and poor illumination [11]. Recent surveys likewise emphasize that multimodal fusion improves robustness precisely in the regimes where camera-only perception is most fragile [25]. Recent fusion methods span feature-level, decision-level, and geometric fusion designs. Deng *et al.* review recent radar-camera object detection advances for autonomous driving and note that the two modalities are complementary in empirical sensing physics and failure characteristics [12]. Xiao *et al.* [13] further show that combining perspective-view and bird’s-eye-view cues improves 3-D object detection performance, highlighting that depth-aware fusion is increasingly central to robust vehicle localization. Even when full radar hardware is unavailable, these studies still motivate a key design principle: *localization should not depend exclusively on the optical channel* when the target application is safety-critical and must operate under glare, rain, or low-light perturbations. At the same time, the specific fusion problem in ADB is more constrained than that in general autonomous driving. The objective is not full-scene 3-D understanding, but reliable continuation of the suppression region for the most relevant oncoming light source over short time horizons. This creates two unique requirements. First, when the camera becomes saturated, the

image-plane coordinates become weakly observed or unobservable. Second, any auxiliary depth signal must fail independently from the camera; otherwise, both modalities collapse under the same glare event. Our design follows the logic of the radar-camera fusion literature, but in a cost-constrained form: we combine image-space detections with a physics-based depth surrogate and switch to a Pseudo-Radar-only observation model during visual dropout. In this sense, our method is not intended to replace full radar-camera fusion, but to instantiate its robustness principle in an ADB-specific and low-cost setting.

A second gap in the literature concerns *how* intelligent headlight systems are evaluated. Traditional automotive lighting studies often rely on photometric bench tests, regulatory compliance analysis, or expensive on-road experiments, whereas computer-vision papers typically stop at image-space metrics. For systems such as ADB, neither perspective is fully sufficient: the former underrepresents perception failure modes, while the latter may ignore the downstream consequences of localization errors on actual beam actuation. Recent validation literature increasingly advocates testbench-based hardware-in-the-loop (HIL), software-in-the-loop (SIL), and vehicle-in-the-loop (VIL) methodologies for ADAS and autonomous-vehicle development [26–28]. Cheng *et al.* survey VIL simulation testing for autonomous vehicles and emphasize its role in bridging pure simulation and costly full-road experiments through repeatable, physically grounded testing setups [26]. More recently, Langer *et al.* demonstrate a VIL framework specifically for front-camera verification using adaptive high beam, underscoring the growing need for reproducible evaluation of perception-dependent lighting functions [27]. Broader recent perspectives on SIL/HIL in automotive testing also argue that closed-loop virtual-physical validation is increasingly necessary as perception, control, and embedded implementation become tightly coupled [28].

These developments strongly support the use of a compact HIL platform in our setting. However, existing VIL/HIL work on lighting is still primarily concerned with functional verification or platform-level testing, and is much less focused on the specific computer-vision failure mode in which glare destroys the observation required to maintain the beam suppression mask. Our HIL framework is designed to address exactly this gap. By coupling the perception stack, the adaptive tracking logic, and a physical LED actuation layer, it enables evaluation at the level that ultimately matters for ADB control: whether the correct suppression zone is maintained in the presence of brief but safety-critical visual collapse. This is also why we later introduce an actuation-level metric rather than relying exclusively on conventional detection scores.

Camera-centric ADB pipelines face a structural limitation that the four strands reviewed above do not jointly address: when the front-facing camera is saturated by intense oncoming headlights, the suppression decision must be sustained without a reliable image-plane observation. Early production ADB designs relied on simple threshold-based headlight detection [20], which is robust under moderate illumination but collapses under blooming and flare. More recent camera-based ADB proposals have improved detection accuracy under normal nighttime conditions, yet headlight-induced saturation remains an underexplored failure mode in the open literature. At the same time, the HIL and VIL validation literature [26, 27] has demonstrated the value of reproducible closed-loop testing for perception-dependent lighting functions,

but existing HIL studies focus primarily on functional verification rather than on the specific failure mode of glare-induced detection collapse. The present work is positioned at the intersection of these gaps: it combines a camera-independent depth prior with adaptive Kalman filtering to sustain actuation continuity during brief visual saturation events, and evaluates the result through an actuation-level metric rather than exclusively in pixel space.

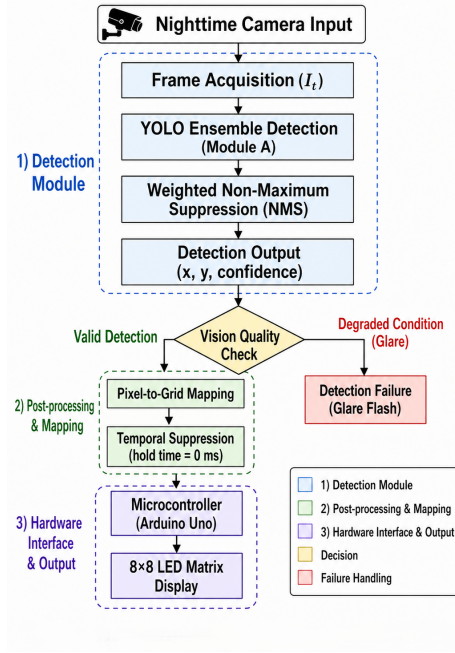
In summary, recent literature has made clear progress in four directions: adaptive headlight control [21, 22], nighttime vehicle detection under challenging illumination conditions [8–10], radar–camera fusion for robust automotive perception [11–13], and HIL/VIL validation for intelligent vehicle systems [26–28]. Yet the intersection of these strands remains underexplored. In particular, the literature still lacks a low-cost, end-to-end framework that addresses *glare-induced detection collapse* in camera-centric ADB, maintains localization through short periods of observation loss, and evaluates the result at the level of discrete headlight actuation rather than only in pixel space. Our work is intended to fill this gap. The next section will describe the proposed AuraBeam framework with a system overview and the detailed modules inside.

3 System Methodology

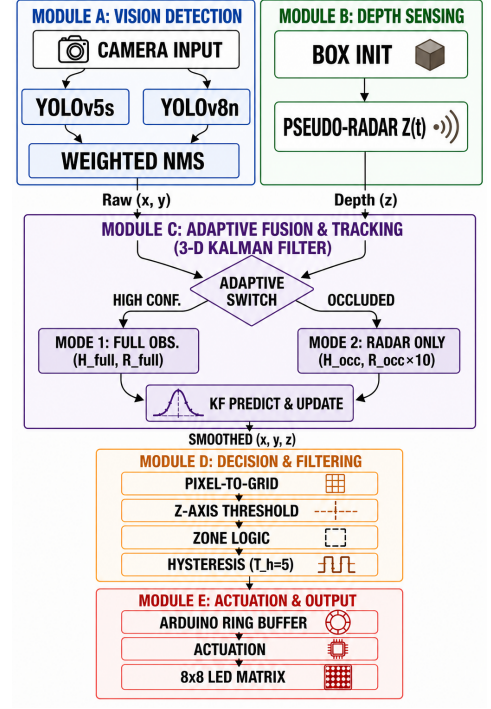
3.1 System Overview

Figure 1 presents the overall AuraBeam pipeline. The system is organized as a closed-loop perception-to-actuation process comprising five modules: (A) an ensemble vision front-end for nighttime vehicle detection, (B) a physics-based depth surrogate referred to as Pseudo-Radar, (C) an adaptive 3-D Kalman filtering module for state estimation under partial observation loss, (D) a zone-mapping and anti-flicker controller that converts the estimated target state into a discrete suppression pattern, and (E) a hardware actuation layer based on an Arduino-driven LED matrix. Unlike a conventional camera-only ADB pipeline, AuraBeam is designed to preserve localization continuity when the image stream becomes unreliable due to blooming, exposure disruption, or severe glare. The detector provides image-plane observations when confidence is sufficient, whereas the state estimator continues propagating the suppression trajectory during short intervals of visual collapse by relying on its internal motion model and the auxiliary depth signal. The output of this estimation stage is then translated into a suppression region on an 8×8 LED grid and sent to the hardware layer for real-time actuation.

To realize the HIL setup, we employ a low-cost peripheral hardware stack consisting of an Arduino Uno R3, an 8×8 LED matrix, a USB camera, and a serial communication interface. The processing unit executes the perception and filtering modules, while the microcontroller serves as the physical actuation layer. As summarized in Table 1, the total peripheral cost is approximately \$23, excluding the host computer. This compact implementation is intended as a reproducible experimental platform rather than a production-grade automotive lighting system. For easier comprehension, Table 2 summarizes the main symbols and notations used throughout this paper.



(a) YOLO-only baseline pipeline.



(b) AuraBeam full system pipeline.

Fig. 1: Architectural comparison between the YOLO-only baseline pipeline and the proposed AuraBeam pipeline.

Table 1: Hardware bill of materials.

Component	Specification	Est. Cost (USD)
Processing unit	PC or laptop with discrete GPU	—
Microcontroller	Arduino Uno R3	\$5
LED matrix	1588BS 8×8 module	\$1
Camera	USB 1080p webcam or dashcam	\$15
Serial interface	USB-A to USB-B cable	\$2
Total		\$23

3.2 Module A: Ensemble YOLO Vision Pipeline

The first stage of AuraBeam performs nighttime vehicle detection using an ensemble of two complementary object detectors. Let I_t denote the input frame at time t . Two models, denoted $\mathcal{M}_1$ and $\mathcal{M}_2$, are executed in parallel on I_t . The first model is specialized for high-glare headlight patterns, whereas the second is trained on a broader nighttime vehicle dataset to improve generalization across diverse scene structures and vehicle appearances. Each model produces a set of detections $\mathcal{D}_i = \{(b_j, s_j)\}_{j=1}^{N_i}$, where

Table 2: Main symbols and notations.

Symbol	Description
I_t	Input frame at time t
$\mathcal{M}_1, \mathcal{M}_2$	Detection models in the ensemble pipeline
$\mathcal{D}_i$	Detection set produced by model $\mathcal{M}_i$
$b_j = [c_x, c_y, w, h]^\top$	Bounding box in center-width-height format
s_j	Confidence score of a detection
$\hat{b}, \hat{s}$	Fused bounding box and fused confidence score
$\mathbf{d}_t$	Image-space observation vector at time t
$\mathbf{x}_k$	3-D Kalman filter state vector at step k
$\mathbf{F}$	State transition matrix
$\mathbf{Q}$	Process noise covariance matrix
$\mathbf{H}_{\text{full}}$	Observation matrix in full-observation mode
$\mathbf{H}_{\text{occ}}$	Observation matrix in occlusion-aware mode
$\mathbf{R}_{\text{full}}$	Measurement covariance in full-observation mode
$\mathbf{R}_{\text{occ}}$	Measurement covariance in occlusion-aware mode
$Z(t)$	Pseudo-Radar depth prior at time t
$\mathcal{B}_t$	Suppression region on the LED grid at time t
$\mathbf{G}_t$	Output LED suppression grid at time t
τ_{NMS}	IoU threshold for Weighted NMS
τ_{conf}	Confidence threshold for observation switching
T_h	Hysteresis hold-time window

$b_j = [c_x, c_y, w, h]^\top$ is a bounding box in center-width-height format and $s_j \in [0, 1]$ is the associated confidence score. Because the two detectors are trained on purposefully different data distributions, their outputs are expected to be complementary, particularly under low-light conditions in which one detector may better preserve light-source sensitivity while the other more effectively captures vehicle shape cues.

To combine the predictions, AuraBeam applies Weighted Non-Maximum Suppression (Weighted NMS). For two overlapping detections (b_a, s_a) and (b_b, s_b) that satisfy $\text{IoU}(b_a, b_b) \geq \tau_{\text{NMS}}$, the fused bounding box $\hat{b}$ and fused confidence $\hat{s}$ are computed as

$$\hat{b} = \frac{w_1 s_a b_a + w_2 s_b b_b}{w_1 s_a + w_2 s_b}, \quad (1)$$

$$\hat{s} = \frac{w_1 s_a + w_2 s_b}{w_1 + w_2}, \quad (2)$$

where $w_1, w_2 \in \mathbb{R}_{>0}$ are detector-specific weights selected on the validation set, and τ_{NMS} is fixed to 0.5. The output of Module A at frame t is the image-space observation vector

$$\mathbf{d}_t = [c_x, c_y, w, h, \hat{s}]^\top \in \mathbb{R}^5. \quad (3)$$

This representation retains both geometric information and a confidence signal that is later used by the adaptive observation logic in the Kalman filter.

Training Configuration

The two detectors are trained independently in order to maximize feature diversity.

Model $\mathcal{M}_1$ (YOLOv5).

Model $\mathcal{M}_1$ was trained on a custom high-glare nighttime dataset containing 5,534 original images. A versioned dataset was created in Roboflow, resulting in 9,934 images after preprocessing and augmentation. The exported dataset was split into 8,800 training images, 831 validation images, and 303 test images. Preprocessing included auto-orientation and resizing to 640×640 . Data augmentation included horizontal flipping, random rotation in the range of -7° to $+7^\circ$, brightness perturbation between -19% and $+19\%$, and Gaussian blur up to 1.5 px. The augmentation pipeline generated two outputs per training example.

Model $\mathcal{M}_2$ (YOLOv8).

Model $\mathcal{M}_2$ was implemented using YOLOv8 and trained on an in-house nighttime driving dataset comprising 6,275 source images across two classes. After excluding unannotated samples, the dataset was partitioned into training, validation, and testing subsets following an approximately 4.4k/1.3k/611 split. The model was trained for 100 epochs at an input resolution of 1024×1024 with batch size 8. Early stopping with patience 30 was used to reduce overfitting. Because nighttime scenes are highly sensitive to sparse luminous structures, the augmentation strategy combined strong spatial augmentation—Mosaic, MixUp, Copy-Paste, scaling, and small-angle rotation—with controlled photometric perturbation. In particular, hue variation was disabled to avoid distorting the red/white light signatures that are important for nighttime vehicle perception.

After training, the ensemble weights w_1 and w_2 in Eqs. (1) and (2) are selected on the validation split to optimize downstream detection performance.

3.3 Module B: Pseudo-Radar Depth Estimation

A key requirement of AuraBeam is to maintain target localization when the visual signal temporarily degrades. Instead of introducing a dedicated physical radar or LiDAR sensor, which would increase hardware cost and integration complexity, we construct a lightweight auxiliary depth signal termed *Pseudo-Radar*. This signal is not intended to replace real range sensing; rather, it provides a simple depth prior that remains independent of instantaneous image brightness and can therefore continue to evolve during visual saturation.

Assuming that the oncoming vehicle follows an approximately constant approach velocity over short time intervals, the estimated range at time t is defined as

$$Z(t) = Z_0 - v_{\text{app}}t, \quad (4)$$

where Z_0 is the initial range estimated at first acquisition and v_{app} is the relative approach velocity. In the experimental implementation, the initial range Z_0 is approximated using a standard pinhole camera model based on a prior of average vehicle width, calculated during the first sequence of high-confidence acquisitions. Furthermore, because monocular depth derivatives are extremely noisy under low-light conditions, the approach velocity v_{app} is not differentiated frame-by-frame. Instead, it is initialized to a nominal closing speed acting as a structural prior. This design

intentionally sacrifices precise dynamic ranging in favor of a stable, noise-free depth progression during short camera dropouts. Since $Z(t)$ is propagated analytically from the initial condition and elapsed time, it is unaffected by camera glare or short-term detector failure.

The approximation in Eq. (4) is accurate only when the relative motion is close to constant velocity. If the true longitudinal acceleration is $a_{\text{real}} \neq 0$, then the actual range evolves as

$$Z_{\text{true}}(t) = Z_0 - v_{\text{app}}t - \frac{1}{2}a_{\text{real}}t^2, \quad (5)$$

and the resulting depth estimation error is

$$\Delta Z(t) = Z(t) - Z_{\text{true}}(t) = -\frac{1}{2}a_{\text{real}}t^2. \quad (6)$$

This error grows quadratically with time. For example, under moderate braking with $a \approx -0.5g$, the absolute depth error reaches approximately 2.45 m after one second of occlusion; under stronger braking with $a \approx -0.9g$, it may reach approximately 4.4 m. Consequently, the Pseudo-Radar module should be interpreted as a low-cost temporal depth surrogate suitable for short dropout intervals, not as a substitute for production-grade radar sensing.

3.4 Module C: Adaptive 3-D Kalman Filter

3.4.1 State Representation and Motion Model

To integrate image-plane localization with the auxiliary depth signal, AuraBeam maintains a six-dimensional state vector

$$\mathbf{x}_k = [c_x, c_y, Z, \dot{c}_x, \dot{c}_y, \dot{Z}]^\top \in \mathbb{R}^6, \quad (7)$$

where (c_x, c_y) denote the image-space centroid of the target, Z denotes the estimated range, and $(\dot{c}_x, \dot{c}_y, \dot{Z})$ are the corresponding image-plane and longitudinal velocities. A constant-velocity state transition model is used:

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad (8)$$

where $\Delta t = 33.3$ ms is the frame interval. The process noise covariance is initialized as

$$\mathbf{Q} = \text{diag}(0.08, 0.08, 0.03, 0.03, 0.03, 0.02), \quad (9)$$

with the first three entries corresponding to (c_x, c_y, Z) and the last three entries corresponding to $(\dot{c}_x, \dot{c}_y, \dot{Z})$. These extremely low process noise values were selected empirically to compel the filter to act as a strong low-pass smoother. Because nighttime

bounding boxes exhibit high spatial variance and abrupt jitter during glare events, the filter is tuned to heavily trust the internal motion model over instantaneous camera measurements. While this strict tuning introduces a minor geometric lag, it is essential for ensuring that the mapped suppression command remains stable across the discrete LED grid, which helps explain why the fixed-observation baseline demonstrates such strong inherent robustness.

3.4.2 Observation Models

AuraBeam operates in two observation modes depending on the detector confidence.

Mode 1: Full observation.

When the detector confidence is sufficiently high, the state is updated using image-plane coordinates and depth. The observation model is

$$\mathbf{H}_{\text{full}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \quad (10)$$

with measurement vector

$$\mathbf{y}_k^{\text{full}} = [c_x, c_y, Z(t)]^\top. \quad (11)$$

The corresponding measurement noise covariance matrix is

$$\mathbf{R}_{\text{full}} = \text{diag}(30.0, 30.0, 0.5), \quad (12)$$

where the first two terms are measured in px^2 and the third term in m^2 .

Mode 2: Occlusion-aware reduced observation.

When the per-frame fused confidence remains below a threshold τ_{conf} for L consecutive frames, the image-plane observation is treated as unreliable and the filter updates only the depth component. In the final protocol, the confidence signal is the best fused detection confidence in the current frame, and the temporal guard is fixed to $L = 2$ consecutive low-confidence frames. The observation model then becomes

$$\mathbf{H}_{\text{occ}} = [0 \ 0 \ 1 \ 0 \ 0 \ 0], \quad (13)$$

with measurement vector

$$\mathbf{y}_k^{\text{occ}} = [Z(t)]. \quad (14)$$

In this mode, the filter relies on the internal motion model to propagate (c_x, c_y) while continuing to anchor the state using the depth prior. The reduced-observation measurement covariance is one-dimensional:

$$\mathbf{R}_{\text{occ}} = [0.5]. \quad (15)$$

Conceptually, this temporally guarded mode switch allows the estimator to preserve short-term localization continuity without trusting unstable image measurements during glare-induced detector collapse, while also avoiding single-frame toggling caused by transient confidence dips. During reduced-observation updates, the image-plane

uncertainty evolves through the prediction covariance rather than through direct (c_x, c_y) measurements.

3.4.3 Recursive Estimation

The prediction step is given by

$$\hat{\mathbf{x}}_k^- = \mathbf{F}\hat{\mathbf{x}}_{k-1}, \quad (16)$$

$$\mathbf{P}_k^- = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^\top + \mathbf{Q}, \quad (17)$$

where $\hat{\mathbf{x}}_k^-$ and $\mathbf{P}_k^-$ denote the prior state estimate and covariance. The Kalman gain is then computed as

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}^\top (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^\top + \mathbf{R})^{-1}, \quad (18)$$

followed by the update equations

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k (\mathbf{y}_k - \mathbf{H}\hat{\mathbf{x}}_k^-), \quad (19)$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_k^-, \quad (20)$$

where $(\mathbf{H}, \mathbf{R}, \mathbf{y}_k)$ are selected according to the current observation mode.

3.4.4 Adaptive Detect-then-Track Procedure

Algorithm 1 summarizes the complete inference loop. At each frame, the system first produces a fused image-space observation from the detector ensemble, then computes the Pseudo-Radar depth, predicts the next target state, evaluates the confidence guard, selects the observation mode, updates the state estimate, and finally maps the filtered localization to the LED control grid.

3.5 Module D: Zone Mapping and Anti-Flicker Scheduling

Pixel-to-Grid Projection

The filtered centroid $(\hat{c}_x, \hat{c}_y)$ is projected onto the 8×8 LED grid as

$$g_{\text{col}} = \left\lfloor \frac{\hat{c}_x}{W_{\text{frame}}} \cdot 8 \right\rfloor, \quad g_{\text{row}} = \left\lfloor \frac{\hat{c}_y}{H_{\text{frame}}} \cdot 8 \right\rfloor, \quad (21)$$

where W_{frame} and H_{frame} are the image width and height. A square suppression region $\mathcal{B}_t$ centered at $(g_{\text{col}}, g_{\text{row}})$ is then formed using a half-width parameter δ measured in LED cells. More broadly, the transformation of continuous sensing geometry into discrete decision units also arises in other camera-oriented sensing systems, where discretization plays an important role in preserving downstream control or coverage behavior [29].

Algorithm 1 AuraBeam detect-then-track loop with adaptive observation switching

Require: Video stream $\{I_t\}$, thresholds τ_{conf} and τ_{NMS} , approach velocity v_{app}

Ensure: LED suppression grid $\mathbf{G}_t$ for each frame

```
1: Initialize  $\hat{\mathbf{x}}_0$ ,  $\mathbf{P}_0$ ,  $\mathbf{Q}$ ,  $\mathbf{R}_{\text{full}}$ , and  $\mathbf{R}_{\text{occ}}$ 
2: Estimate initial range  $Z_0$  from the first valid target acquisition
3: for each frame  $t = 1, 2, \dots$  do
4:   Run  $\mathcal{M}_1$  and  $\mathcal{M}_2$  on  $I_t$  to obtain  $\mathcal{D}_1$  and  $\mathcal{D}_2$ 
5:   Fuse detections using WeightedNMS to obtain  $\mathbf{d}_t$ 
6:   Compute depth prior  $Z(t) = Z_0 - v_{\text{app}}t$ 
7:   Predict  $\hat{\mathbf{x}}_t^-$  and  $\mathbf{P}_t^-$  using Eqs. (16) and (17)
8:   if  $\hat{s}_t \geq \tau_{\text{conf}}$  or fewer than  $L$  consecutive low-confidence frames have occurred
     then
9:     Set  $\mathbf{H} \leftarrow \mathbf{H}_{\text{full}}$ ,  $\mathbf{R} \leftarrow \mathbf{R}_{\text{full}}$ , and  $\mathbf{y}_t \leftarrow \mathbf{y}_t^{\text{full}}$ 
10:   else
11:     Set  $\mathbf{H} \leftarrow \mathbf{H}_{\text{occ}}$ ,  $\mathbf{R} \leftarrow \mathbf{R}_{\text{occ}}$ , and  $\mathbf{y}_t \leftarrow \mathbf{y}_t^{\text{occ}}$ 
12:   end if
13:   Update  $\hat{\mathbf{x}}_t$  and  $\mathbf{P}_t$  using Eqs. (18)–(20)
14:   Map  $(\hat{c}_x, \hat{c}_y, \hat{Z})$  to the LED grid to obtain  $\mathbf{G}_t$ 
15:   Transmit  $\mathbf{G}_t$  to the actuation layer
16: end for
```

Depth-Dependent Suppression Policy

The suppression mode is modulated according to the estimated target range $\hat{Z}$. When the target is far away, no suppression is applied. At intermediate range, a dimmed suppression mode is used. At close range, the selected LED region is fully turned off. The thresholds used in this work are summarized in Table 3.

Table 3: Depth-dependent glare suppression policy.

Estimated range	LED state	Duty cycle
$\hat{Z} \geq 60$ m	No suppression	100%
$30 \leq \hat{Z} < 60$ m	Partial suppression	40%
$\hat{Z} < 30$ m	Full suppression	0%

Hysteresis-Based Anti-Flicker Control

Direct frame-by-frame switching of LED cells can produce visible flicker when the estimated suppression region oscillates near cell boundaries. To stabilize actuation, we apply a temporal hold mechanism whereby a cell remains suppressed if it belonged to the recent suppression history. Formally,

$$\mathbf{G}_t[r, c] = \text{OFF} \quad \text{if} \quad \exists \tau \in [t - T_h, t] \text{ such that } (r, c) \in \mathcal{B}_\tau, \quad (22)$$

where T_h is the hold time expressed in frames. In all experiments, we set $T_h = 5$, corresponding to approximately 167 ms at the operating frame rate. This hysteresis mechanism reduces visible ghosting and prevents rapid toggling of adjacent cells during short-term localization jitter.

3.6 Module E: Hardware Actuation Layer

The final suppression pattern is transmitted to an Arduino Uno, which scans the LED matrix using row-column multiplexing at 200 Hz. This refresh rate is well above the critical flicker fusion threshold of the human visual system and thus ensures stable visual output. Communication from the host computer uses a 115200-baud serial link with the packet format

`BOX:<col_start>:<col_end>:<row_start>:<row_end>`

The serial receiver is implemented using an interrupt-driven ring buffer so that packet decoding does not block the timer interrupt responsible for LED scanning. As a result, the actuation layer remains stable even when the upstream perception pipeline exhibits variable inference latency.

In summary, AuraBeam integrates a dual-model nighttime detector, a lightweight depth surrogate, an adaptive 3-D state estimator, and a discrete actuation controller within a single closed-loop HIL framework. The overall design is intended to evaluate a specific failure mode of camera-centric ADB—namely the loss of stable suppression under headlight saturation—while remaining sufficiently simple and reproducible for controlled experimentation.

4 Evaluation Protocol

This section defines the pre-specified protocol used for the final evaluation. The protocol is intentionally narrower than the full AuraBeam design space. Rather than re-optimizing all modules simultaneously, it isolates one question: how adaptive observation switching behaves relative to a fixed 3-D observation model when the detector ensemble, tracker family, and hold-time controller are held constant. The goal is therefore not to demonstrate that B4 universally outperforms B3, but to establish a defensible scenario-by-scenario boundary for when switching helps, when it is neutral, and when it fails.

4.1 Implementation Details

All final experiments use the complete AuraBeam pipeline described in Section 3, but with a single fixed detector and tracker stack. The perception front-end is fixed to the dual-model YOLOv5/YOLOv8 ensemble fused by Weighted NMS. The tracking layer is fixed to the six-state KF3D model together with the Pseudo-Radar depth surrogate. The actuation layer maps the filtered target state onto an 8×8 LED grid and applies the hold-time controller of Section 3.5 with $T_h = 5$ frames before transmission to the Arduino-based HIL platform.

The final comparison contains only two configurations:

- **B3: KF3D fixed observation.** The tracker always uses the full observation model when detections are available.
- **B4: KF3D adaptive switching.** The tracker falls back to depth-only updates when the best fused detection confidence is below $\tau_{\text{conf}} = 0.35$ for at least two consecutive frames.

The perception front-end models were trained using cloud-based accelerators: the primary YOLOv5 detector was trained on the Kaggle platform utilizing dual NVIDIA T4 GPUs, while the secondary YOLOv8 detector was trained on Google Colab using a single NVIDIA T4 GPU. All hardware-in-the-loop experiments, tracking algorithms, and evaluation protocols are implemented in the Python programming language. The closed-loop inference and system evaluations were executed locally on an HP Victus machine equipped with an AMD RyzenTM 5 7535HS processor. We have provided the complete source code to ensure the experiments are fully reproducible. The switching policy is the legacy `best_conf` policy, and no scenario-specific retuning is performed after the protocol is fixed. Each scenario is replayed five times (`run_01` to `run_05`) under the same offline configuration. Under this fixed replay setting, aggregate actuation metrics are deterministic across the five runs; repeated runs are therefore used primarily as a reproducibility check and to capture runtime variability, not as independent stochastic trials.

4.2 Evaluation Metrics

The evaluation uses three levels of metrics. First, trajectory-level metrics quantify the temporal stability of the estimated target path. Second, control-level metrics quantify whether the final LED suppression pattern correctly covers the opposing headlight region. Third, runtime metrics quantify the computational overhead of the comparison. Because ADB is ultimately an actuation problem, the control-level metrics are treated as primary and the others as supporting evidence.

4.2.1 Primary Control-Level Metric

For a matrix headlight controller, the terminal output is a set of discrete LED cells; a small image-space error may therefore map to the wrong cell being suppressed, resulting in residual glare or unnecessary darkening. A metric defined directly over the LED grid is accordingly required. The GSSR metric is introduced to address this gap:

$$\text{GSSR} = \frac{1}{N} \sum_{t=1}^N \mathbf{1} \left[\text{IoU}_{\text{grid}} \left(\hat{\mathcal{B}}_t, \mathcal{B}_t^* \right) \geq 0.5 \right], \quad (23)$$

where $\hat{\mathcal{B}}_t$ is the set of LED cells predicted by the system to require suppression at time t , and $\mathcal{B}_t^*$ is the ground-truth set of cells corresponding to the actual headlight footprint of the oncoming vehicle on the 8×8 grid. Grid-level IoU is defined as:

$$\text{IoU}_{\text{grid}}(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (24)$$

In the final analysis, the primary comparison uses an *occlusion-window* version of the metric, in which the evaluation is restricted to manually annotated degradation intervals. This yields the three primary metrics reported later in the results section: *occlusion GSSR*, *occlusion missed glare rate*, and *occlusion false darkening rate*. For the negative controls, which contain no designated degradation interval, the full-sequence metrics are reported instead to verify that the protocol does not destabilize normal nighttime operation.

4.2.2 Supplementary Metrics

False Darkening Rate (FDR): The proportion of frames in which the system suppresses at least one LED cell when no opposing vehicle is present:

$$\text{FDR} = \frac{1}{N_0} \sum_{t: \mathcal{B}_t^* = \emptyset} \mathbf{1}[\hat{\mathcal{B}}_t \neq \emptyset]. \quad (25)$$

This metric quantifies unnecessary darkening, which degrades the host driver’s forward visibility. [30]

Missed Glare Rate (MGR): The proportion of frames in which the system fails to suppress any LED cell when an opposing vehicle is present:

$$\text{MGR} = \frac{1}{N_1} \sum_{t: \mathcal{B}_t^* \neq \emptyset} \mathbf{1}[\hat{\mathcal{B}}_t = \emptyset]. \quad (26)$$

MGR captures total suppression failure (i.e., complete target loss). However, because Eq. (26) only checks if *any* LED cell is suppressed—and thus does not penalize spatial misalignment when the wrong cells are darkened—it cannot fully represent system safety. Consequently, the grid-aware GSSR remains the primary and most stringent safety metric in this framework.

Trajectory RMSE:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{t=1}^N \left\| \begin{bmatrix} \hat{c}_{x,t} \\ \hat{c}_{y,t} \end{bmatrix} - \begin{bmatrix} c_{x,t}^* \\ c_{y,t}^* \end{bmatrix} \right\|_2^2}, \quad (27)$$

where $(c_{x,t}^*, c_{y,t}^*)$ denotes the ground-truth centroid coordinates.

Jitter Reduction (JR%):

$$\text{JR} = \left(1 - \frac{\sigma_{\hat{c}}^2}{\sigma_d^2} \right) \times 100\%, \quad (28)$$

where $\sigma_{\hat{c}}^2$ and σ_d^2 are the centroid position variances of the Kalman Filter output and the raw YOLO output, respectively.

In the results section, RMSE and JR% are used to analyze tracking quality; GSSR, FDR, and MGR are used to analyze the full control outcome; and the occlusion-conditioned versions of these metrics are used as the primary basis for the B3/B4

comparison. This separation is intentional: the central claim is about actuation robustness during targeted degradation, not about universal detector superiority.

4.3 Dataset, Scenarios, and Baseline Configurations

Evaluation is performed on a manually curated *Golden Test Set* reserved exclusively for final benchmarking and not used for detector training. Pixel-level bounding boxes are annotated in Roboflow and then projected onto LED grid cells via Eq. (21) to obtain actuation-level ground truth. The final benchmark is intentionally organized around five failure-oriented scenarios that serve three roles in the argument: negative controls, targeted degradation cases, and a limitation case. Table 4 summarizes the two comparison configurations, and Table 5 summarizes the five scenarios. Earlier exploratory studies considered broader detector and control ablations, but they are not used as the basis of the final claim. This structure supports a narrow interpretation of the later results. Because B3 and B4 share the same detector ensemble, KF3D backbone, and actuation controller, any difference between them can be attributed to the adaptive observation policy rather than to detector replacement, tracker family changes, or controller redesign.

Table 4: Evaluation configurations used in the experimental protocol

Group	Configuration	Purpose
B3	KF3D + fixed observation	Fixed-baseline configuration
B4	KF3D + adaptive switching	Adaptive-switching configuration

Golden Test Set Construction

A *Golden Test Set* of 953 video frames is constructed via manual annotation using Roboflow. Pixel-level bounding boxes are subsequently projected onto LED grid cells via Eq. (21). Table 5 summarizes the five evaluation scenarios, while Table 6 details the frame counts and primary visual challenges for each sequence. Sequences are selected to represent adversarial edge cases that induce failure in camera-only ADB systems:

- **Mountain curve at night (negative control).** A curved mountain-road sequence acquired at night, used to verify that the adaptive logic does not degrade a difficult but non-targeted control case.
- **Urban night with street lighting (negative control).** A nominal nighttime urban sequence used to verify that the protocol preserves stable operation when no targeted degradation is present.
- **Foggy night (targeted degradation).** Diffuse glare and bloom caused by fog stress the confidence-triggered switching mechanism over a sustained degradation interval.
- **Lightning disturbance at night (targeted degradation).** Abrupt exposure disruption creates a short but severe degradation interval and tests whether the

Table 5: Five-case protocol used in the final evaluation

Scenario	Role	Purpose
Mountain curve at night	Negative control	Verify no control failure on a difficult curved-road night sequence
Urban night with street lighting	Negative control	Verify no unnecessary degradation under nominal urban nighttime conditions
Foggy night	Targeted degradation	Test sustained confidence collapse under fog-induced bloom
Lightning disturbance at night	Targeted degradation	Test short abrupt exposure disruption
Rainy night	Limitation case	Expose the heavy-rain failure boundary

adaptive logic can preserve actuation continuity without harming the surrounding frames.

- **Rainy night (limitation case).** Heavy rain and specular reflection noise probe a regime in which confidence-based switching may fail or overreact, and are treated explicitly as a limitation test rather than a guaranteed success case.

Table 6: Frame counts and dominant challenges in the five-case protocol

Scenario	Challenge Type	Frames	Primary Difficulty
Mountain curve at night	Curved-road nighttime control	76	Difficult nighttime visibility on a mountain curve
Urban night with street lighting	Urban nighttime control	204	Baseline operation under street illumination
Rainy night	Specular reflection noise	219	Road reflections and lens water droplets
Foggy night	Strong light scattering	215	Blooming and diffuse glare obscure vehicle features
Lightning disturbance at night	Abrupt illumination transient	239	Exposure disruption causes short multi-frame instability
Total		953	



Fig. 2: Representative Golden Test Set frames. (a) Mountain curve at night; (b) Urban night with street lighting; (c–d) Rainy night; (e) Foggy night; (f–g) Lightning disturbance at night.

All results reported in Section 5 are computed on the same Golden Test Set with identical post-processing and actuation settings. Scenario-wise reporting is retained because the dominant failure mechanisms differ substantially across low-light controls, rain, scattering, and abrupt exposure disruption. No scenario-specific threshold tuning is performed at test time. This fixed-protocol evaluation is intended to expose genuine robustness rather than gains attributable to per-scenario optimization.

5 Experimental Results

This section first reports supporting ablations for the perception, tracking, and control components. These ablations characterize the broader AuraBeam design, but they are not used as the primary basis for the final claim. The main claim is evaluated by the fixed-protocol B3/B4 comparison reported in the negative-control, targeted-degradation, and limitation-case subsections.

5.1 Perception Ablations (A1–A4)

Table 7 reports the detector-side ablations A1–A4. The perception metrics are averaged over the five-case Ultralytics evaluation at IoU 0.5. For the two single-model

rows, mAP@0.5, precision, and recall are computed directly from the Ultralytics validation summaries. For the ensemble rows, the same metrics are computed after NMS or weighted NMS fusion using the selected confidence and fusion-IoU settings for each case. The detector-level metrics show a clear but not complete alignment with the downstream control metrics. The primary YOLOv5 detector has a stronger five-case mean mAP@0.5 (0.306 versus 0.243) and also leads on the stricter single-model mAP50–95 metric (0.106 versus 0.076). It also has slightly higher precision and recall than the secondary detector. However, the secondary detector remains useful because it is competitive in rainy and night-control conditions. Standard NMS fusion increases recall to 0.426 but lowers precision to 0.353, indicating that it recovers additional candidates at the cost of more false positives. Weighted NMS keeps nearly the same recall (0.425) while improving precision to 0.374 and mAP@0.5 to 0.292, which explains why A4 is retained as the perception front-end for the full AuraBeam pipeline.

Table 7: Perception ablations A1–A4

ID	Detector Configuration	mAP@0.5	Precision	Recall	GSSR	Occl. GSSR	Notes
A1	Primary detector only	0.306	0.429	0.369	43.51%	0.00%	YOLOv5 baseline
A2	Secondary detector only	0.243	0.417	0.306	39.07%	22.61%	YOLOv8 baseline
A3	Ensemble NMS (ours)	0.274	0.353	0.426	56.15%	22.61%	Higher recall, lower precision
A4	Ensemble weighted (ours)	0.292	0.374	0.425	56.32%	22.61%	Proposed fusion

The low five-case mean should be interpreted with this scenario mixture in mind. It is not caused by uniformly poor detector behavior across all cases. Under the weighted-NMS ensemble, the normal-night sequence reaches mAP@0.5 of 0.543 with precision 0.510 and recall 0.652, and the urban night-control sequence reaches mAP@0.5 of 0.346 with precision 0.469 and recall 0.492. The lightning-disturbance sequence also remains usable at the aggregate level, with mAP@0.5 of 0.359 and recall 0.623, although its head-light class has many false positives. The mean is pulled down primarily by the targeted degradation cases, especially fogging (mAP@0.5 of 0.009 and recall 0.056) and rain (mAP@0.5 of 0.205 and recall 0.300). Thus, the detector metrics indicate a highly scenario-dependent front-end: adequate in clearer nighttime scenes, but fragile under dense scattering and saturation. To provide a clearer breakdown of this variance,

Table 8 details the performance of the weighted-NMS ensemble across each individual scenario.

Table 8: Per-scenario detection metrics for the weighted-NMS ensemble (A4). The five-case mean is pulled down primarily by fogging and rain; clearer nighttime scenes achieve substantially higher scores.

Scenario	mAP@0.5	Precision	Recall
Mountain curve at night	0.543	0.510	0.652
Urban night with street light	0.346	0.469	0.492
Lightning disturbance at night	0.359	0.294	0.623
Rainy night	0.205	0.438	0.300
Foggy night	0.009	0.157	0.056
Five-case mean	0.292	0.374	0.425

5.2 Tracking and Observation Ablations (B1–B4)

Table 9: Tracking and observation ablations B1–B4

ID	Tracking Configuration	RMSE	JR%	GSSR	FDR	MGR	Occl. GSSR	Occl. FDR
B1	Raw detection / no KF	122.32	0.00	56.32%	6.69%	19.98%	22.61%	1.75%
B2	KF2D	56.16	2.68	71.10%	5.80%	1.20%	48.15%	7.99%
B3	KF3D fixed observation	56.16	2.68	91.10%	8.54%	0.37%	84.60%	15.40%
B4	KF3D adaptive switching	56.49	3.03	90.82%	8.81%	0.37%	77.19%	22.81%

Table 9 details the tracking-side comparison from B1 to B4. This subsection is kept distinct from the main B3/B4 discussion below so that numbers can be analyzed incrementally. The results reveal that the main tracking benefit is obtained in two stages. First, replacing raw detections with KF2D smoothing substantially stabilizes the image-plane trajectory: RMSE drops from 122.32 px in B1 to 56.16 px in B2, while full-sequence GSSR rises from 56.32% to 71.10% and MGR falls from 19.98% to 1.20%. This indicates that temporal filtering is essential for maintaining suppression continuity when the detector output is noisy or intermittent. Second, moving

from KF2D to the fixed-observation KF3D model improves the actuation outcome even though the mean image-plane RMSE remains unchanged under this protocol. B3 raises full-sequence GSSR from 71.10% to 91.10% and occlusion-window GSSR from 48.15% to 84.60%. The gain therefore does not come primarily from reducing centroid error; it comes from using the depth-aware state to produce a more reliable LED-grid suppression command. This result supports the use of 3-D state estimation for ADB control, where small differences in range and projected position can change which LED cells should be suppressed. The adaptive observation variant B4 preserves most of the full-sequence performance of B3, with only a small GSSR decrease from 91.10% to 90.82% and the same full-sequence MGR of 0.37%. However, the occlusion metrics expose a clear limitation: occlusion GSSR decreases from 84.60% to 77.19%, while occlusion FDR increases from 15.40% to 22.81%. Thus, the adaptive switch does not mainly fail by losing the target completely; instead, it tends to over-suppress or mis-align the LED-grid command during degraded intervals. For the current protocol, B3 is therefore the stronger aggregate tracking baseline, and B4 should be interpreted as a scenario-dependent fallback mechanism rather than a uniformly superior observation policy.

5.3 Control and Switching Ablations (C1–C4)

Table 10: Control and switching ablations C1–C4

ID	Control Configuration	GSSR	FDR	MGR	Occl. GSSR	Occl. FDR	Occl. MGR	Switch count	CmdChg
C1	Direct / no hold-time	90.45%	8.35%	1.20%	77.19%	22.81%	0.00%	9.4	17.44%
C2	Hold-time only	90.82%	8.81%	0.37%	77.19%	22.81%	0.00%	9.4	17.26%
C3	KF3D fixed obs + hold	91.10%	8.54%	0.37%	84.60%	15.40%	0.00%	0	17.35%
C4	KF3D adapt switch + hold	90.82%	8.81%	0.37%	77.19%	22.81%	0.00%	9.4	17.26%

Table 10 outlines the control-side comparison C1–C4. The purpose is to observe how hardware scheduling logic impacts the final outcome. The measured ablations separate three effects. First, perception fusion raises the full-sequence GSSR from 43.51% for the primary detector to 56.32% for weighted fusion, but the occlusion-window GSSR is identical for A2–A4. The perception-side gain is therefore visible in the overall control outcome, while the hard occlusion cases remain governed by the tracking and observation policy. Second, the tracking ablation produces the largest system-level gain. Moving from raw detections to KF2D reduces RMSE from 122.32 px

to 56.16 px and lowers MGR from 19.98% to 1.20%. Moving from KF2D to the fixed-observation KF3D configuration keeps the same mean RMSE under this protocol but raises GSSR from 71.10% to 91.10%, indicating that the benefit is expressed most clearly at the LED-grid actuation level rather than only as a centroid-error reduction. Third, the control ablation shows a safety trade-off rather than a one-sided improvement. Enabling hold-time reduces MGR from 1.20% to 0.37%, but increases FDR from 8.35% to 8.81%. C1, C2, and C4 all use the adaptive observation policy and therefore show the same mean switch count of 9.4 across the five scenarios, whereas C3 is the fixed observation reference and has zero switches. The command-change rate decreases only slightly, from 17.44% in C1 to 17.26% in C2/C4, so this metric should be interpreted as a small stability signal rather than a large effect.

5.4 Negative Controls and Targeted Degradation Cases

To provide a comprehensive overview of the experimental findings, Figure 3 summarizes the five-case evaluation protocol alongside the primary actuation results. The upper panel outlines the visual challenges associated with each sequence, while the lower panel previews the comparative GSSR performance of the fixed-observation baseline (B3) against the adaptive observation fallback (B4). A detailed breakdown of these scenarios follows below.

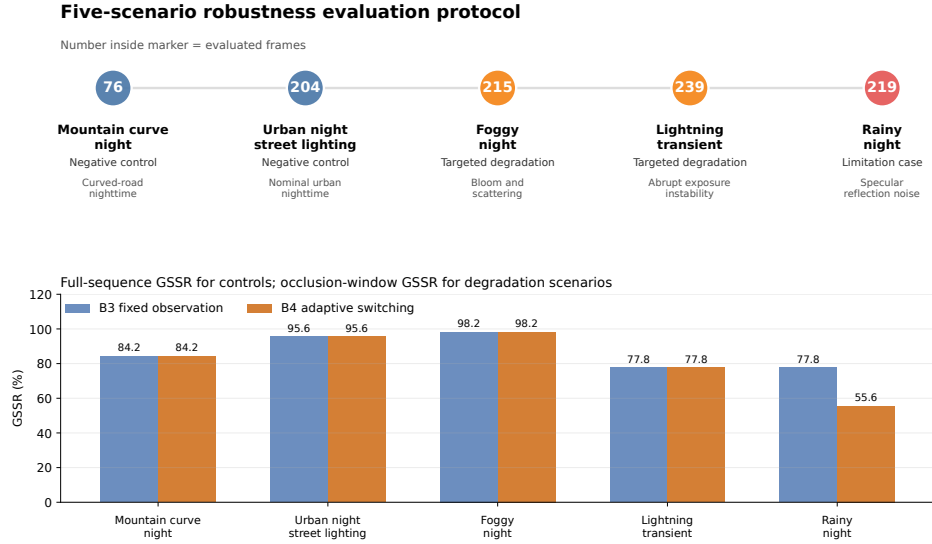


Fig. 3: Five-scenario robustness evaluation protocol. The upper panel summarizes the evaluated frame counts, scenario roles, and dominant visual challenges for the two negative controls, two targeted degradation scenarios, and the rainy-night limitation case. The lower panel compares B3 and B4 using full-sequence GSSR for the control scenarios and occlusion-window GSSR for the degradation and limitation scenarios.

Table 11: Negative controls: full-sequence actuation metrics (% , mean of five replays)

Scenario	Role	B3		B4		Shared MGR	B4 switches
		GSSR	FDR	GSSR	FDR		
Mountain curve at night	Negative control	84.21	15.79	84.21	15.79	0.00	2
Urban night with street lighting	Negative control	95.59	4.41	95.59	4.41	0.00	0

Table 11 reports the full-sequence actuation metrics for the two negative controls. These sequences are used to verify that the adaptive policy does not damage normal or near-normal operation. The control cases confirm that the adaptive policy neither improves nor degrades the aggregate actuation outcome relative to the fixed KF3D baseline. In the urban night sequence with street lighting, B4 never enters the depth-only mode. In the mountain-curve night sequence, B4 switches only twice on average, yet the aggregate metrics remain identical to B3. This behavior is important for the main claim: the protocol preserves the controls, but the controls alone do not justify any claim of superiority for adaptive switching.

Table 12: Primary comparison on degradation intervals (% , mean of five replays)

Scenario	Role	B3			B4			
		GSSR	MGR	FDR	GSSR	MGR	FDR	Switches
Foggy night	Targeted degradation	98.25	0.00	1.75	98.25	0.00	1.75	35
Lightning disturbance at night	Targeted degradation	77.78	0.00	22.22	77.78	0.00	22.22	1
Rainy night	Limitation case	77.78	0.00	22.22	55.56	0.00	44.44	9

Table 12 reports the primary metrics, namely the occlusion-window actuation metrics restricted to the manually annotated degradation intervals, with two findings follow directly. First, adaptive switching is *active* in the intended degradation cases. In the foggy-night sequence, B4 enters the reduced observation mode 35 times per replay on average, spending 16.28% of frames in depth-only mode. In the lightning-disturbance sequence, the trigger is far rarer (one switch, 0.42% of frames), reflecting a short disruption window rather than a sustained collapse. Second, this activity does not translate into a blanket aggregate advantage. In the foggy-night sequence, B4 matches B3 exactly on the primary occlusion metrics. In the lightning-disturbance sequence, B4 again matches B3 exactly. These two cases therefore support a narrower interpretation: adaptive switching can preserve actuation performance in selected degradation

cases, but the final results do not support the stronger claim that B4 consistently outperforms B3. The foggy-night case is useful for visualizing how this behavior unfolds inside the pipeline. Figure 4 first presents a qualitative three-stage view of the fallback mechanism. Under normal camera conditions, the detector and depth prior provide a full observation. When fogging and headlight glare cause visual collapse, the image-plane detection becomes unreliable, and the tracker switches to a depth-only update. The Kalman filter then propagates the target trajectory using the retained motion state and the Pseudo-Radar depth prior.

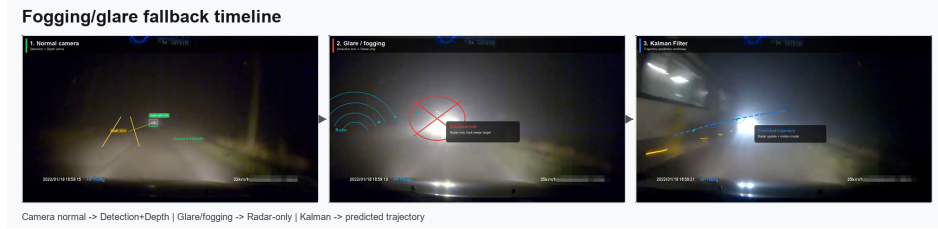


Fig. 4: Three-frame qualitative illustration of the fogging/glare fallback sequence.

Foggy-night switching behavior

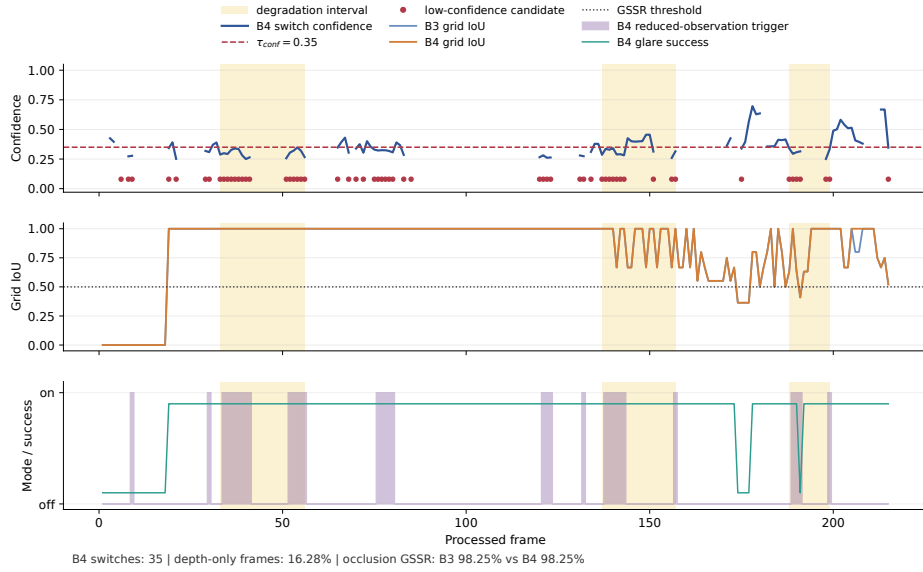


Fig. 5: Foggy-night switching behavior. The timeline shows the confidence signal, shaded degradation interval, B4 depth-only activation markers, and LED-grid actuation quality during the same failure mode illustrated qualitatively in Fig. 4.

To complement the qualitative sequence, Figure 5 reports the corresponding frame-level signals. The confidence trace shows when the switching guard becomes active, while the grid-IoU and glare-success traces verify that the reduced-observation updates preserve the LED-grid suppression outcome during the annotated windows.

5.5 Limitation Case and Stability

Rainy-night limitation case

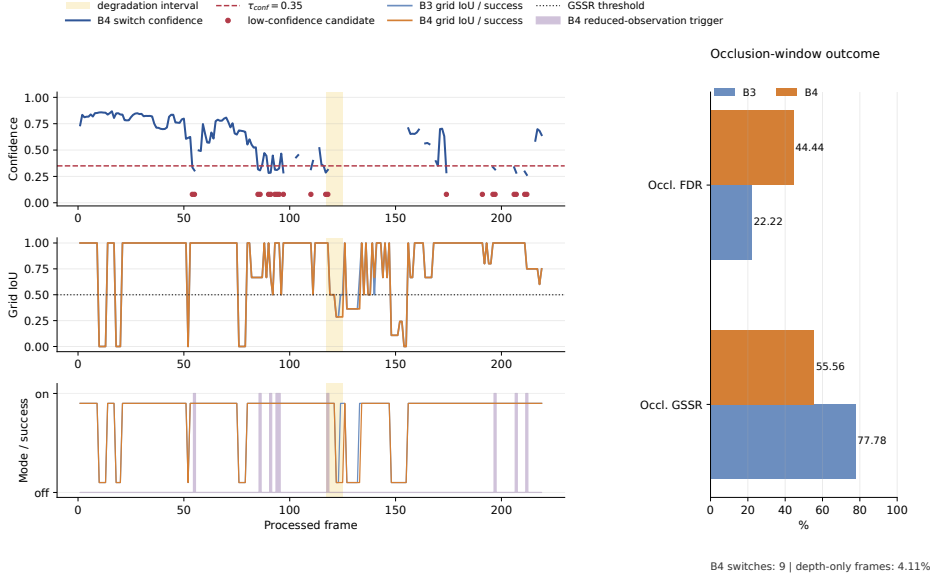


Fig. 6: Rainy-night limitation case. The time-series panels show B4 switch confidence, grid IoU, reduced-observation activations, and glare-suppression success during the rain sequence. The summary bars highlight the boundary of the adaptive policy: B4 reduces occlusion-window GSSR relative to B3 while increasing occlusion false darkening.

The principal limitation case is the rainy-night sequence. Under heavy rain and specular reflection noise, B4 falls from 77.78% to 55.56% occlusion GSSR, and its occlusion false darkening rate rises from 22.22% to 44.44%. The missed glare rate remains at 0.00% for both B3 and B4, but the rain sequence nonetheless establishes an important limitation: confidence-based switching is not a universal replacement for fixed observation. In this regime, the adaptive policy triggers nine times per replay on average, yet the resulting trajectory is less well-behaved at the LED-grid level than the fixed-observation baseline. This misalignment occurs because rapid confidence-triggered toggling during sustained specular noise disrupts the temporal continuity required by the hysteresis controller. To illustrate this failure mode, Figure 6 presents

a detailed timeline of the system’s behavior during the rainy-night degradation interval, highlighting how the confidence-triggered fallback leads to unnecessary command changes and reduced suppression success. The B3/B4 comparison introduces only a marginal computational difference. Averaged over the five scenarios, B3 yields a mean total latency of 277.65 ms per frame, while B4 yields 282.12 ms per frame. The adaptive logic therefore introduces only a small measured runtime difference relative to the detector stack. Trajectory-level metrics likewise show only minor differences between B3 and B4 under the pre-specified protocol. For example, the full-sequence RMSE values are nearly identical in the mountain-curve night sequence (36.14 vs. 36.13 px), the urban night sequence with street lighting (16.49 vs. 16.49 px), the foggy-night sequence (136.27 vs. 136.95 px), and the lightning-disturbance sequence (11.71 vs. 11.72 px). The decisive evidence for the main claim, therefore, derives from the occlusion-conditioned actuation metrics rather than from large tracking-RMSE gaps.

5.6 Protocol-Level Paired Checks

Table 13: Protocol-level paired comparisons for the supporting ablation studies. Wilcoxon signed-rank p -values are computed over paired scenario-run entries.

Comparison	Metric	Left mean	Right mean	Δ	Wilcoxon p	Interpretation
A1 $\rightarrow$ A4	GSSR $\uparrow$	43.51	56.32	+12.81	7.9×10^{-5}	Ensemble weighted improves the control-level perception baseline.
A3 $\rightarrow$ A4	GSSR $\uparrow$	56.15	56.32	+0.17	0.025	Weighted fusion is slightly higher, but the effect size is small.
B1 $\rightarrow$ B2	RMSE $\downarrow$	122.32	56.16	-66.16	1.1×10^{-5}	KF2D substantially reduces centroid error.
B2 $\rightarrow$ B3	GSSR $\uparrow$	71.10	91.10	+20.00	1.1×10^{-5}	KF3D gives the largest actuation-level robustness gain.
C1 $\rightarrow$ C2	CmdChg $\downarrow$	17.44	17.26	-0.19	0.025	Hold-time slightly reduces command changes, lowering MGR.
B3 $\rightarrow$ B4	Occl. GSSR $\uparrow$	84.60	77.19	-7.41	0.025	Adaptive switching is worse on the occlusion-window aggregate.
C3 $\rightarrow$ C4	CmdChg $\downarrow$	17.35	17.26	-0.09	0.025	Command changes decrease but occlusion GSSR also decreases.

To summarize the direction and consistency of the supporting ablation comparisons, we report Wilcoxon signed-rank checks over paired scenario-run entries in Table 13. These values are *not* used for independent stochastic-trial inference: the five replays are deterministic offline executions of the same fixed scenarios, so the resulting p -values do not correspond to independent random draws. They should be read exclusively as protocol-level paired consistency checks that verify whether each design change produces a directionally stable effect within the fixed evaluation protocol.

Perception improvement (A1–A4): The protocol-level paired comparison indicates a consistent improvement in system control outcomes when the ablation perception module is integrated. The GSSR increased from 43.51% in A1 to 56.32% in A4. At the detector level, weighted NMS also improves the ensemble mAP@0.5 from 0.274 to 0.292 and precision from 0.353 to 0.374, while recall remains nearly unchanged (0.426 to 0.425). Incorporating confidence-weighted fusion (A4) therefore slightly exceeded standard NMS (A3) in both detector-level precision and aggregate control outcome, increasing mean GSSR from 56.15% to 56.32%.

Tracking and state estimation (B1–B3): Applying 2D KF smoothing (B2) substantially reduced the tracking positioning error compared to the baseline (B1). The RMSE decreased from 122.32 px to 56.16 px. Furthermore, transitioning from a 2D state to a 3D state estimation (B3) provided the largest actuation-level gain in this protocol, increasing the *GSSR* from 71.10% to 91.10%.

Control scheduling (C1 vs. C2): The addition of a temporal hold-time logic (C2) improved the safety-oriented missed-glare behavior compared to direct frame-by-frame mapping (C1), reducing MGR from 1.20% to 0.37%. The Box Command Change Rate also showed a small decrease from 17.44% to 17.26%, consistent with a modest stabilization effect.

Occlusion protocol (B3 vs. B4): When evaluating the adaptive switching mechanism during occlusion intervals (*occlusion GSSR*), the success rate dropped from 84.60% (B3) to 77.19% (B4). This result identifies a boundary of the current confidence-based switching design rather than supporting a universal improvement claim.

Control stability (C3 vs. C4): The adaptive configuration (C4) resulted in a minor reduction in command-change rate relative to the fixed-observation baseline C3. The Box Command Change Rate changed from 17.35% to 17.26%, while the occlusion GSSR decreased from 84.60% to 77.19%. This confirms that the smoother command sequence does not automatically imply better occlusion-window actuation.

5.7 Discussions and Limitations

The quantitative results support the role of robust tracking, state estimation, and actuation hysteresis, while also defining a clear boundary for the current adaptive switching policy.

Perception Ensembling: Utilizing an ensemble pipeline improved the glare suppression baseline relative to the primary detector alone. At the detector level, YOLOv5 is the stronger single model on the five-case mean, with mAP@0.5 of 0.306, precision of 0.429, and recall of 0.369, compared with 0.243, 0.417, and 0.306 for the YOLOv8/model.ai detector. This advantage is most visible in the normal-night and

lightning-disturbance cases. The secondary detector is nevertheless complementary: it performs better in the rainy sequence and in one night-control sequence, where its higher precision reduces spurious detections. After fusion, standard NMS increases mean recall to 0.426 but reduces precision to 0.353, whereas weighted NMS keeps a similar recall (0.425) while improving precision to 0.374 and mAP@0.5 to 0.292. Weighted NMS (A4) also yields a slightly higher mean GSSR (56.32%) than standard ensemble NMS (A3, 56.15%). The A3–A4 gap is small, so the stronger conclusion is not that weighted fusion dominates every case, but that the ensemble family improves the control outcome compared with the weaker single-detector baselines.

The per-case detector metrics also explain why the mean detector scores appear modest despite acceptable behavior in several individual scenes. The weighted-NMS ensemble performs reasonably in clearer or less severely degraded cases: normal night reaches mAP@0.5 of 0.543, precision of 0.510, and recall of 0.652; the urban night-control case reaches mAP@0.5 of 0.346; and the lightning-disturbance case reaches mAP@0.5 of 0.359 with recall of 0.623. However, fogging remains the hardest detector-level condition: the ensemble mAP@0.5 is only 0.009 and recall is 0.056, even after fusion. Rain is also difficult, with weighted-NMS mAP@0.5 of 0.205 and recall of 0.300. These low-scoring degradation cases pull down the five-case mean and show that the front-end is highly scenario dependent rather than uniformly weak. The thunder/-lightning sequence also exposes class imbalance: rear-light recall is high (0.812 under weighted NMS), but head-light precision remains very low (0.086), producing many false candidates. These results show that the perception front-end alone is insufficient for stable ADB control under degradation, motivating the subsequent Kalman filtering, depth prior, and hold-time control modules.

Trade-off between mAP and Recall: The comparison between the primary baseline (A1) and the weighted ensemble (A4) reveals an important trade-off between standard detection metrics and downstream control performance. While A1 achieves a higher mean mAP@0.5 (0.306), its recall is lower (0.369). The ensemble (A4) sacrifices some mAP (0.292) to boost recall (0.425). In the context of ADB, missing a headlight (false negative) causes dangerous glare, whereas an extra detection (false positive) merely causes unnecessary dimming. Thus, trading mAP for recall is highly favorable for system safety, directly driving the substantial increase in the downstream GSSR from 43.51% to 56.32%.

Efficacy of 3D State Estimation: The most substantial performance gains were observed in the tracking ablation (Group B). The basic 2D smoothing successfully stabilized the bounding box positioning, cutting the pixel error by more than half. More importantly, upgrading to a 3D state representation (B3) directly translated to a 20% absolute gain in glare suppression success. This confirms the hypothesis that understanding the spatial depth and 3D trajectory of the target is essential for maintaining lock under harsh lighting conditions.

Control Stability and Hysteresis Logic: The integration of a hysteresis hold-time block (C2) directly tackles safety-critical continuity. By retaining the suppression state for a small frame buffer, the system bridges micro-interruptions in perception and reduces MGR from 1.20% to 0.37%. This comes with a conservative increase in FDR from 8.35% to 8.81%, meaning the controller suppresses extra LED cells more

often in exchange for fewer missed-glare frames. The command-change-rate reduction is small, from 17.44% to 17.26%, so the main benefit of hold-time in this protocol is missed-glare reduction rather than a large visible reduction in command switching.

Value and Boundary of Adaptive Switching: The comparison between B3 and B4 exposes the main value of the adaptive observation study. B4 is not valuable because it dominates the fixed-observation baseline; it is valuable because it reveals when a confidence-triggered fallback can preserve actuation and when it can damage the downstream command. In the foggy-night and lightning-disturbance cases, B4 matches B3 during the occlusion windows, showing that the reduced-observation mode can maintain the suppression outcome during selected visual degradation events. In the rain case, however, B4 lowers occlusion GSSR from 77.78% to 55.56% and raises occlusion FDR from 22.22% to 44.44%. At the aggregate level, B4 lowers mean occlusion GSSR from 84.60% to 77.19% and raises occlusion FDR from 15.40% to 22.81%, while occlusion MGR remains at 0.00%. The failure mode is therefore not complete target loss; it is over-darkening or grid-level misalignment during degraded intervals. This negative result supports the need for actuation-aware evaluation: image-space detection metrics or trajectory RMSE alone would not fully expose the downstream false-darkening behavior. Future iterations must re-evaluate the switching trigger and the smoothing logic used during partial camera degradation, especially under heavy rain and specular reflection noise.

This study is intentionally limited to a controlled, low-cost machine-vision research platform. First, the evaluation follows a five-scenario protocol designed to separate negative controls, targeted degradation cases, and a limitation case; therefore, it does not cover the full diversity of real-world nighttime traffic, weather, road geometry, or multi-vehicle interactions. Second, the Pseudo-Radar module is a physics-based depth surrogate rather than a physical radar sensor, so the current results should be interpreted as evidence for the value of a camera-independent depth prior, not as validation against real millimeter-wave radar or LiDAR ground truth. Furthermore, the Pseudo-Radar relies on an initial image-space acquisition to establish the baseline range (Z_0). If the target is saturated immediately upon entering the field of view, this monocular depth surrogate cannot be initialized, representing a structural limitation of camera-only setups. Third, the HIL platform uses a coarse 8×8 LED actuator and the proposed GSSR metric measures LED-grid coverage rather than calibrated illuminance, so the reported results should not be interpreted as production-level photometric or regulatory compliance. Finally, projecting 2D bounding boxes onto a discrete LED grid serves as a reliable geometric approximation for the GSSR metric, though it may not fully capture the complex photometric scattering of physical headlights in real-world environments. These limitations define the boundary of the contribution: AuraBeam is intended as an actuation-aware machine-vision framework for studying degradation behavior and fallback policies, rather than as a production-ready ADB system.

6 Conclusion

This paper introduced AuraBeam, a closed-loop framework for nighttime vehicle localization and glare suppression in camera-centric adaptive driving beam systems. By

combining ensemble detection, a Pseudo-Radar depth prior, adaptive 3-D Kalman filtering, and LED-based Hardware-in-the-Loop evaluation, the study examined whether adaptive observation switching improves actuation robustness under glare-induced visual degradation. The final results do not support a universal advantage of adaptive switching. Instead, its effect is scenario-dependent: it preserves the negative controls, matches the fixed-observation baseline in foggy-night and lightning-disturbance cases, and degrades performance in the rainy-night limitation case. These findings clarify the boundary of the proposed strategy and emphasize that robustness in ADB depends not only on detection quality, but also on stable temporal estimation and actuation-aware evaluation. Future work will extend the framework toward more reliable switching policies, richer sensing, and broader real-world validation.

Declarations

The authors declare no conflict of interest, particularly no financial and personal relationships with other people or organizations that could inappropriately influence (bias) this work.

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